

Program : Diploma in Textile Technology	
Course Code : 5069B	Course Title: Garment CAD Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To impart a clear idea on product development and detailed process of garment development using CAD.
- To develop an understanding on application of computers in pattern making, pattern grading and marker making for garment manufacture.

Course Prerequisites:

Topic	Course code	Course name	Semester
Computing fundamentals		Introduction to IT system	1
Fashion Design & Apparel Manufacture		FDAM	5

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate various tools used in Garment CAD	11	Applying
CO2	Practice the drafting procedure through CAD software	10	Applying
CO3	Apply the pattern grading application using CAD software	11	Applying
CO4	Develop marker for various garments	10	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate various tools used in Garment CAD		
M1.01	Learning of different tools used for drafting - Setting the screen for drafting by using the appropriate tools like Limits, unfold mirror, flip line curves, saving open nest and close nest.	3	Applying
M1.02	Practicing of drafting, editing and saving- Draw Free Hand Sketches using any relevant software.	2	Applying
M1.03	Draw Fashion figure of gents, ladies & children's using any relevant software	3	Applying
M1.04	Draw Striped, Checked Printed and One Way Fabric Design using any relevant software.	3	Applying
CO2	Practice the drafting procedure through CAD software		
M2.01	Design library for garment details (Skirts, Sleeves, Cuffs, Collars, and Pockets)	1	Applying
M2.02	Design flat sketches along with stitch specification for the following: Children – Girls (A-line & yoke frock), Boys (shirt & shorts)	3	Applying
M2.03	Design flat sketches along with stitch specification for the following: Adults – Women's (Top, Skirt, gown), Men's (Shirt, Kurtha, Trouser)	3	Applying
M2.04	For the above create spec sheets, cost sheets for each garment using Fashion Studio software / Photoshop / Corel Draw.	3	Applying
	Lab Exam -1	1.5	

CO3	To understand the pattern grading application through CAD software		
M3.01	Grade the basic bodice block for female one size up and one size down as per given size chart and measurement. (Size 32", 34", 36" on fabric width 45")	3	Applying
M3.02	Grade the Skirt block of all pattern one size up and one size down as per given size chart and measurement. (Size 32", 34", 36" on fabric width 45")	2	Applying
M3.03	Grade the Shirt block of all pattern one size up and one size down as per given size chart and measurement.(Shirt size:40", 42", 44" on fabric width 36")	3	Applying
M3.04	Grade the trouser block of all pattern one size up and one size down as per given size chart and measurement.(Trouser size:32", 34", 36"on fabric width 56")	3	Applying
CO4	To Develop marker on various garments		
M4.01	Prepare the marker and print the same for the following Garments with different fabric texture 1. Yoke Frock 2. Shirt 3. Salwar Kameez 4. Trousers 5. Any coats	4	Applying
M4.02	Compare the manual and computer grading in connection with time and labour.	3	Applying
M4.03	Practicing of calculating the material requirement by using layout.	3	Applying
	Series Test -2	1.5	

Text / Reference:

T/R	Author/Book Title
R1	Taulkdar- Garment Technology- Wood head publishing Ltd
R2	NIFT - Garment Manufacture - New Delhi
R3	Carr.H - The technology of Clothing manufacture
R4	Berry N - Dress pattern designing
R5	Mary Mathew- Practical Clothing Construction

Online Resources:

Sl.No	Website Link
1	http://www.nptel.ac.in/courses
2	http://freetexbooks.blogspot.in/2011/10/handbook-of-weaving.html
3	https://www.youtube.com/
4	http://textilelearner.blogspot.in
5	https://www.textronic.com